

Deep Learning Systems Analysis Report: CIFAR-10 Image Classification

Baseline CNN vs. Dropout-Regularized CNN

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Report Overview

This report addresses an image classification problem: predicting which of 10 object categories a 32x32 color photograph belongs to. The dataset used is CIFAR-10 (Krizhevsky, 2009), a standard computer vision benchmark. A Convolutional Neural Network (CNN) was implemented in PyTorch as a baseline model, and a second CNN differing by exactly one change (added dropout regularization) was trained and compared against it.

Dataset and Task Description

CIFAR-10 consists of 60,000 real, hand-labeled color photographs (50,000 for training, 10,000 for testing), evenly split across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck (Krizhevsky, 2009). Each image is a 32x32 pixel, 3-channel (RGB) array. This directly supports a supervised image classification task: the input is the pixel array and the target is the integer class label. The training data was further split into 45,000 training and 5,000 validation images so training progress could be monitored on unseen data without touching the official 10,000-image test set, which was reserved for final evaluation only.

Model Architecture and Design Decisions

The baseline model is a compact CNN with two convolution-ReLU-max pooling blocks followed by a fully connected classification head. The first block applies a 3x3 convolution expanding the 3 input channels to 32 feature maps, followed by ReLU and 2x2 max pooling (32x32 -> 16x16 spatial resolution); the second block expands 32 channels to 64 and pools again (16x16 -> 8x8). This progressive pattern of shrinking spatial resolution while growing channel depth is a standard convolutional design, allowing early layers to detect simple local patterns (edges, color blobs) and later layers to combine them into more complex features using a manageable number of parameters (LeCun, Bengio, & Hinton, 2015). The flattened 64x8x8 feature map feeds into a 128-unit fully connected layer with ReLU, then a final 10-unit linear layer producing raw class scores. ReLU was used throughout because it is computationally cheap and avoids the vanishing-gradient issues associated with saturating activations like sigmoid or tanh (LeCun, Bengio, & Hinton, 2015). The model was trained with cross-entropy loss (the standard choice for multi-class classification with integer labels) and the Adam optimizer, which adapts its learning rate per parameter and generally converges faster than plain stochastic gradient descent without heavy manual tuning (Kingma & Ba, 2015).

Experimental Comparison

The experimental model is identical to the baseline in every respect except one: a dropout layer ($p = 0.5$) was inserted between the two fully connected layers in the classifier head. This isolates a regularization change rather than mixing it with an architecture or hyperparameter change, so any difference in results can be attributed specifically to dropout. Dropout was chosen because it is a simple, well-established technique for reducing overfitting: by randomly deactivating a fraction of units on each training pass, it prevents units from co-adapting too closely to specific training examples and encourages the network to learn more redundant, generalizable representations (Srivastava, Hinton, Krizhevsky, Sutskever, & Salakhutdinov, 2014). This was a meaningful comparison to make because the baseline model showed early signs of overfitting (a widening train/validation accuracy gap), making dropout a directly relevant intervention to test rather than an arbitrary change.

Results and Interpretation

By the final epoch, the baseline model reached 85.5% training accuracy but only 71.3% validation accuracy, a 14.2 percentage-point gap, and its validation loss began increasing after epoch 6 even as training loss kept falling -- a clear overfitting signature. The dropout model reached only 73.0% training accuracy but 72.1% validation accuracy, a gap of under 1 percentage point, with validation loss decreasing smoothly through all 8 epochs. On the held-out test set, the dropout model slightly outperformed the baseline (71.9% vs. 70.7% accuracy; macro-F1 0.718 vs. 0.704), despite having lower training accuracy. This is the trade-off dropout is designed to produce: it sacrifices some capacity to fit the training data in exchange for a model that generalizes better to unseen data. The cost of this change was slower training convergence -- the dropout model's training accuracy climbed more gradually and may not yet have reached its own performance ceiling within the same 8-epoch budget.

Limitations and Risks

The models were trained for only 8 epochs on a relatively small (compared to modern production vision models) CNN, so absolute accuracy in the low-to-mid 70% range reflects a deliberately modest baseline rather than a competitive result; both models likely have headroom for improvement with more epochs, more capacity, or data augmentation. CIFAR-10's 32x32 resolution also discards fine detail that a higher-resolution dataset would retain, which caps how much visual nuance any model trained on it can learn regardless of architecture. On the evaluation side, only one random train/validation split and one random weight initialization were used for each model, so the reported performance difference between baseline and dropout, while consistent with dropout's known behavior, is a single run rather than an average over multiple seeds and could shift somewhat under different random seeds.

Ethical and Responsible Use

Image classification systems built on architectures like this one carry a documented risk of uneven accuracy across subgroups when the training data does not represent all relevant populations equally; in commercial computer vision systems, this has been shown to produce significantly higher error rates for some demographic groups than others (Buolamwini & Gebru, 2018). While CIFAR-10 classifies inanimate objects and animals rather than people, the same underlying risk applies to any CNN trained on an unrepresentative dataset and then deployed on a more diverse real-world input distribution: performance measured on a curated benchmark like CIFAR-10 does not guarantee similar performance on images collected under different conditions (lighting, camera quality, geographic or cultural context). A responsible deployment of any CNN built this way would require evaluating it on data that reflects the actual population and conditions it will be used on, not only the benchmark's held-out test set.

Future Improvements

With more time and compute, a natural next step would be training for more epochs with a learning-rate schedule and comparing additional regularization strategies (for example, data augmentation such as random crops and flips, or batch normalization) rather than dropout alone. A deeper or wider architecture, or a well-known pretrained CNN backbone fine-tuned on CIFAR-10, would likely close a meaningful part of the remaining gap to state-of-the-art CIFAR-10 results. Repeating both the baseline and dropout experiments across multiple random seeds and reporting mean and variance of test accuracy would also make the comparison more statistically robust than the single-run comparison presented here.

References

- Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research (Conference on Fairness, Accountability, and Transparency)*, 81, 77-91.
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.
- Krizhevsky, A. (2009). Learning multiple layers of features from tiny images (Technical Report). University of Toronto.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436-444. <https://doi.org/10.1038/nature14539>
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1), 1929-1958.